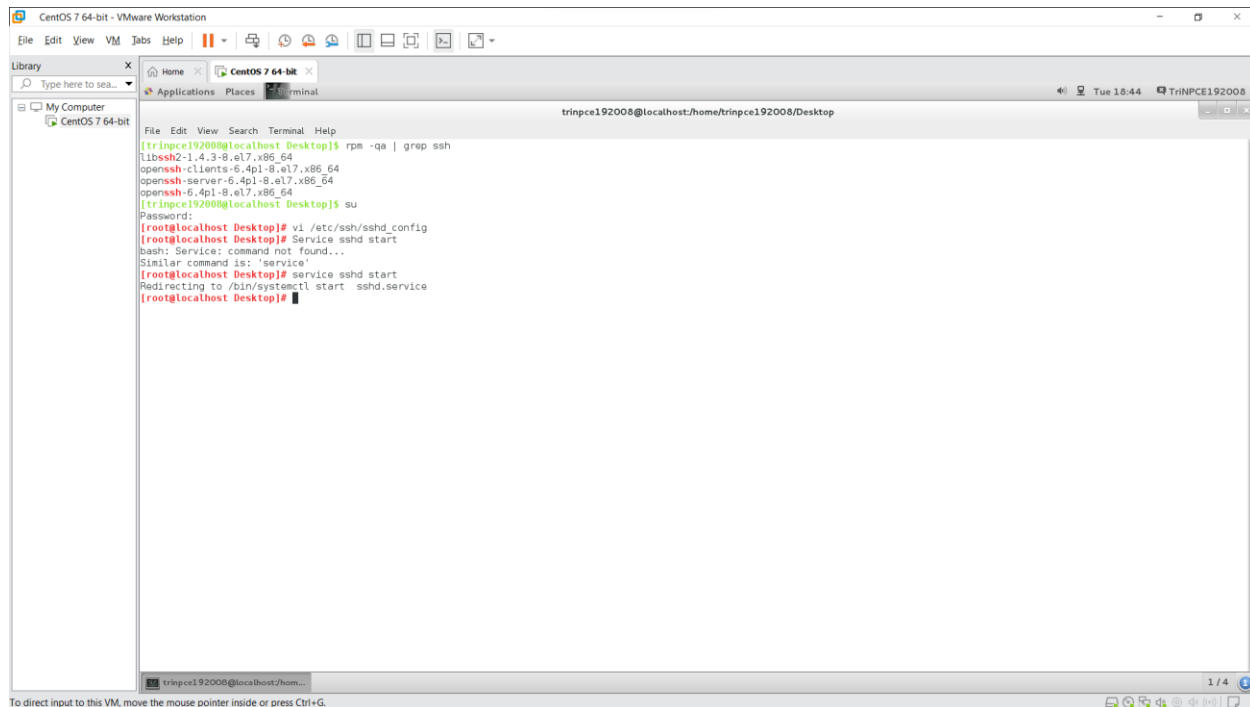


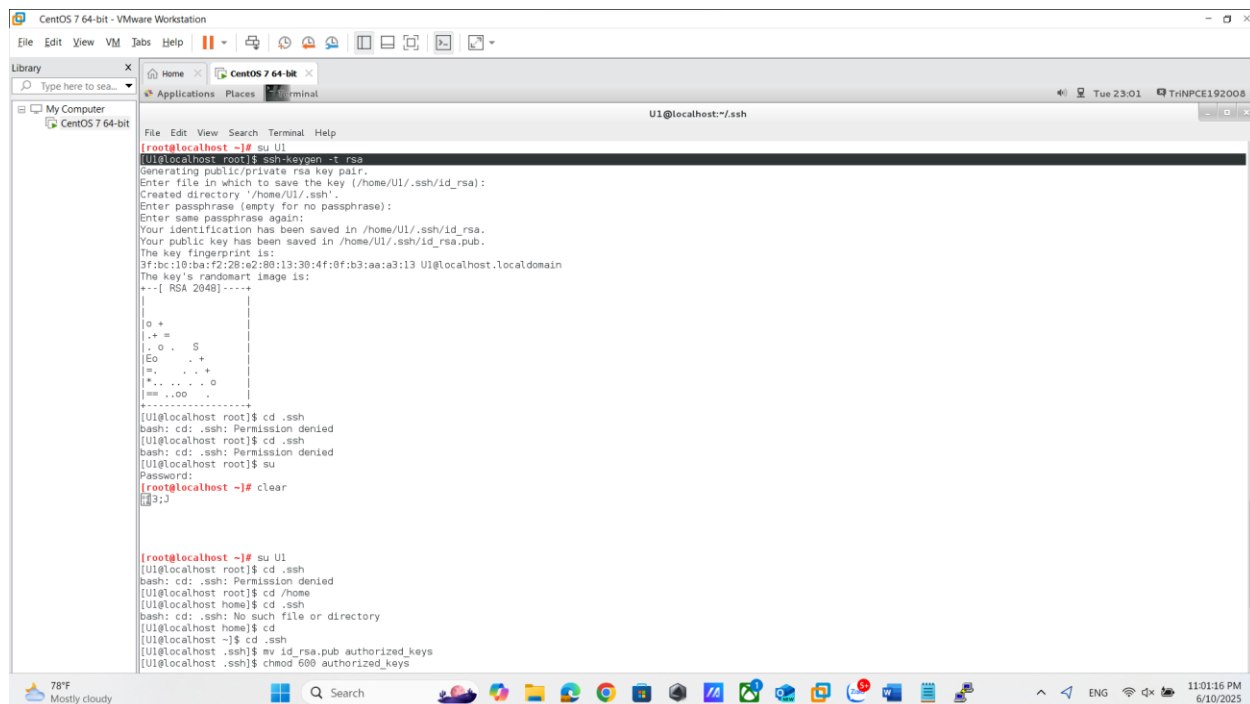
NGUYỄN PHẠM TRÍ LAB 4

SSH:



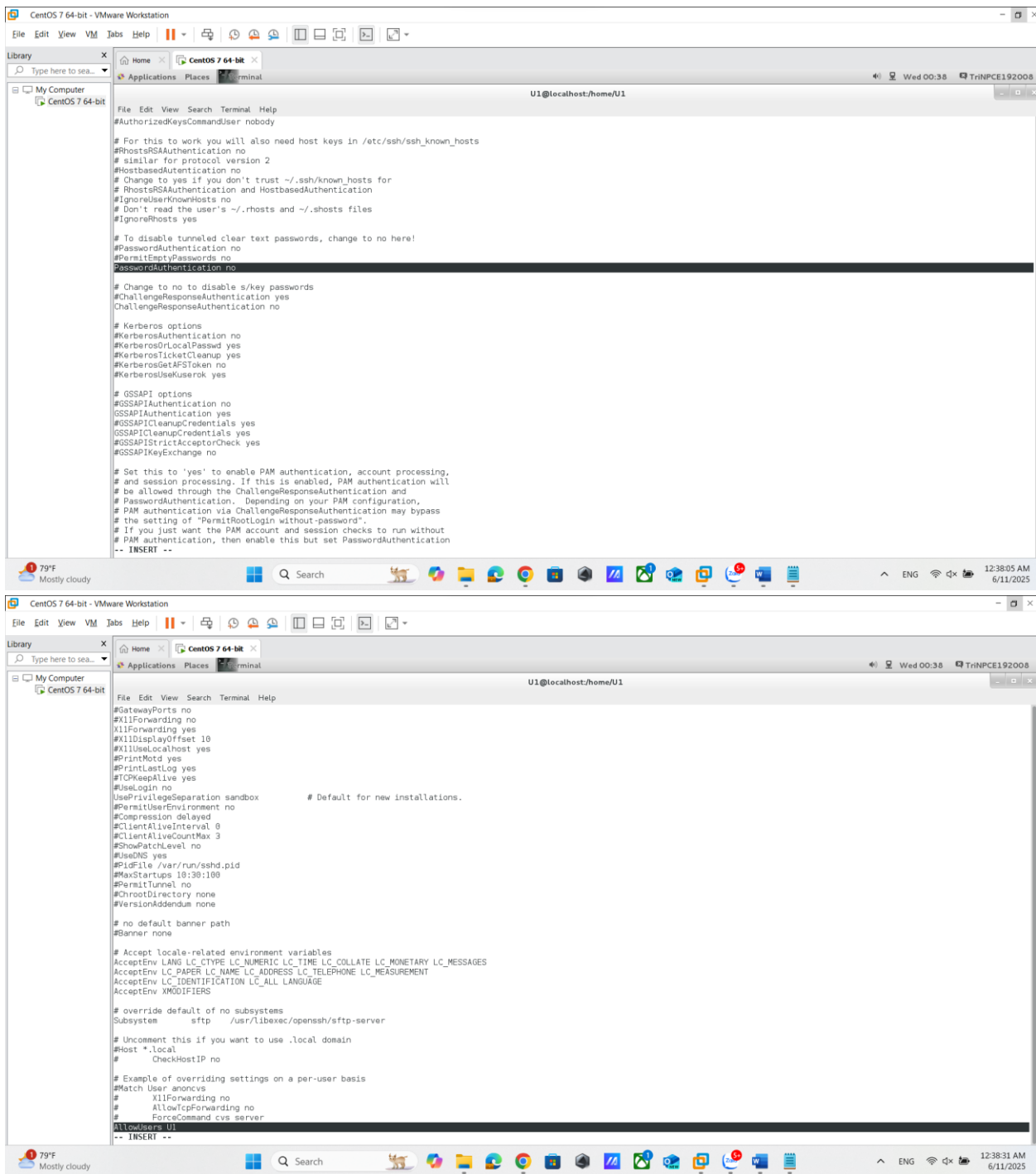
```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
CentOS 7 64-bit
trimpce192008@localhost/home/trimpce192008/Desktop
File Edit View Search Terminal Help
[trimpce192008@localhost Desktop]$ rpm -qa | grep ssh
libssh2-1.4.3-8.el7.x86_64
openssh-clients-6.4p1-8.el7.x86_64
openssh-server-6.4p1-8.el7.x86_64
openssh-6.4p1-8.el7.x86_64
[trimpce192008@localhost Desktop]$ su
Password:
[root@localhost Desktop]# vi /etc/ssh/sshd_config
[root@localhost Desktop]# Service sshd start
bash: Service: command not found...
Similar command is: 'service'
[root@localhost Desktop]# service sshd start
Redirecting to /bin/systemctl start sshd.service
[root@localhost Desktop]#
```

ảnh trên: kiểm tra ssh đã được cài chưa

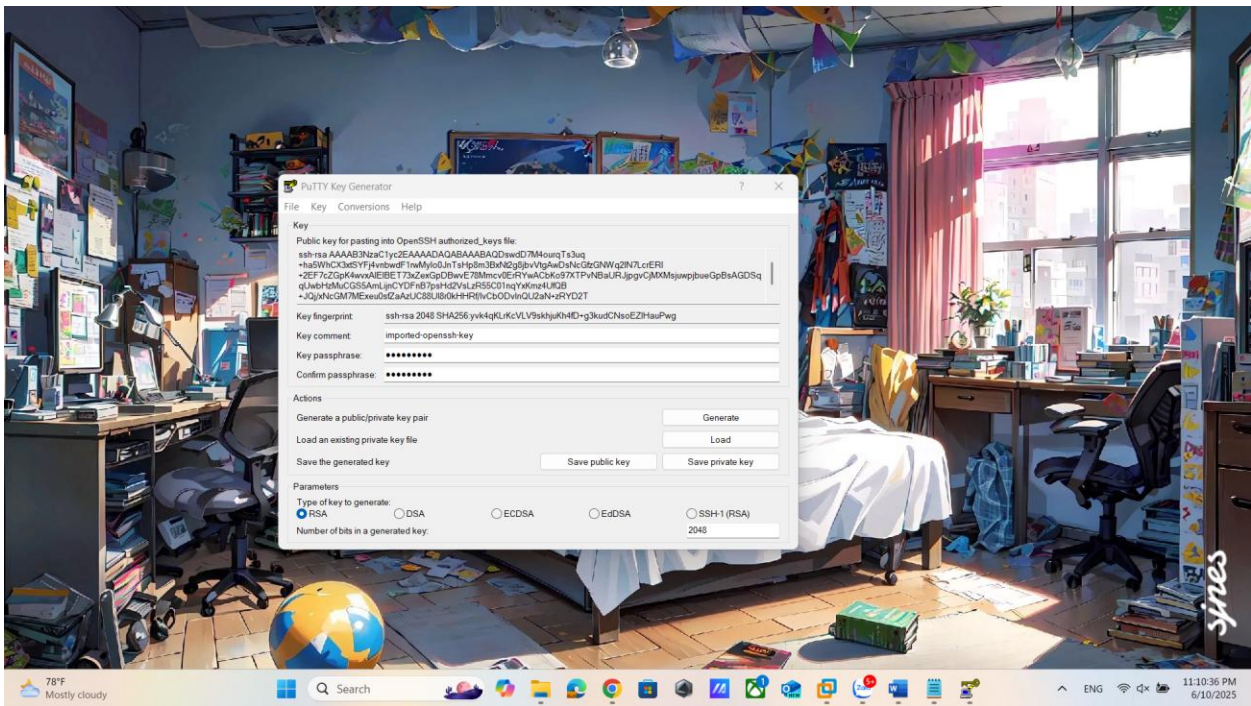


```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
CentOS 7 64-bit
U1@localhost:~/ssh
File Edit View Search Terminal Help
[root@localhost ~]# su U1
[U1@localhost root]$ ssh-keygen -t rsa
Generating public/private rsa key pair.
Enter file in which to save the key (/home/U1/.ssh/id_rsa):
Created directory '/home/U1/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/U1/.ssh/id_rsa.
Your public key has been saved in /home/U1/.ssh/id_rsa.pub.
The key fingerprint is:
3fbc:10:ba:f2:28:c2:90:13:30:4f:0f:b3:aa:a3:13 U1@localhost.localdomain
The key's randomart image is:
+--[ RSA 2048 ]-----+
|
|o+
|.o+
|Eo
|=.
|*..
|=.o
+-----+
[U1@localhost root]$ cd .ssh
bash: cd: .ssh: Permission denied
[U1@localhost root]$ cd .ssh
bash: cd: .ssh: Permission denied
[U1@localhost root]$ su
Password:
[root@localhost ~]# clear
[3:]

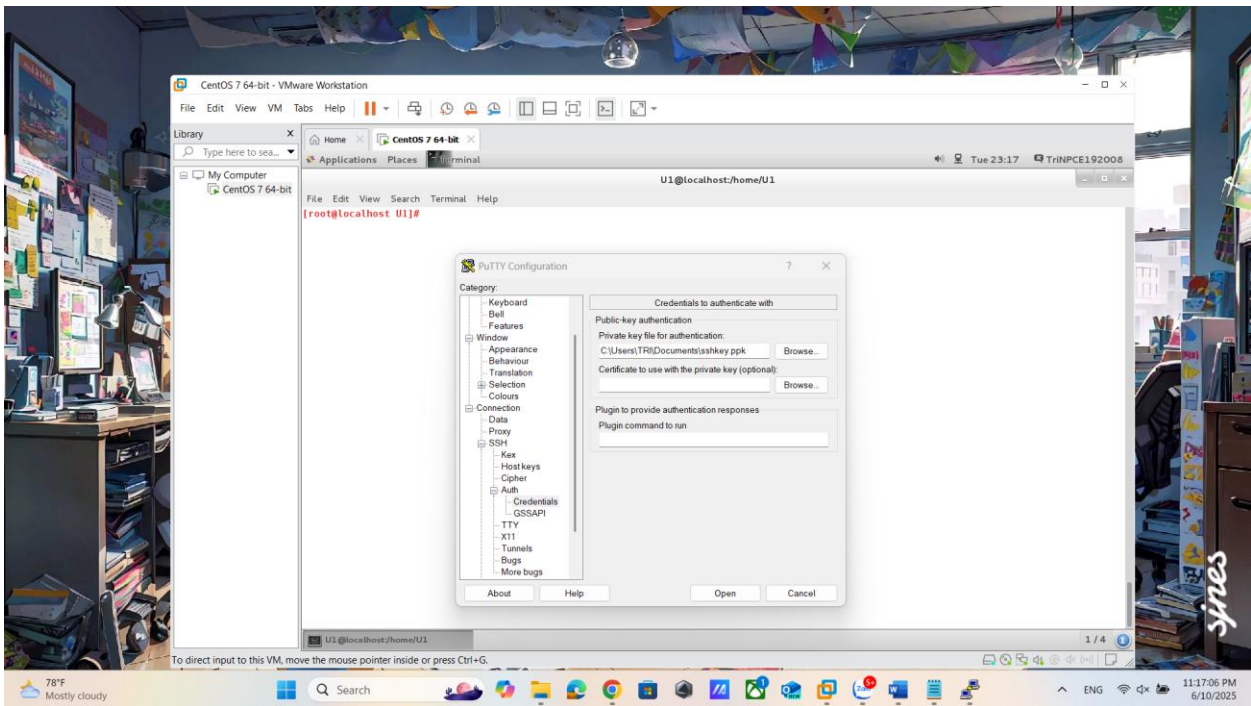
[root@localhost ~]# su U1
[U1@localhost root]$ cd .ssh
bash: cd: .ssh: Permission denied
[U1@localhost root]$ cd /home
[U1@localhost home]$ cd .ssh
bash: cd: .ssh: No such file or directory
[U1@localhost home]$ cd
[U1@localhost ~]$ cd .ssh
[U1@localhost .ssh]$ mv id_rsa.pub authorized_keys
[U1@localhost .ssh]$ chmod 600 authorized_keys
```

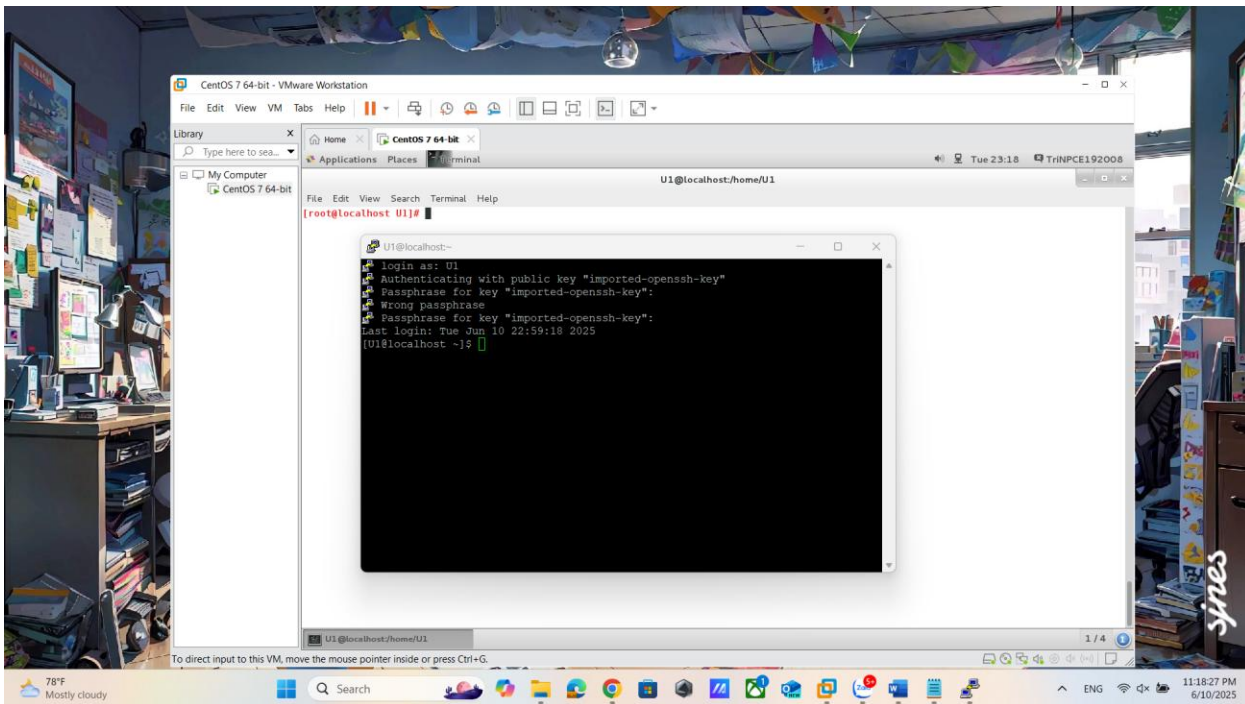
2 ảnh trên: PasswordAuthentication no (đổi lại thành no) và AllowUsers U1 (chỉ cho phép users U1 SSH)



ảnh trên: load file private key.ppk bằng puttygen và lưu lại thành sshkey.ppk



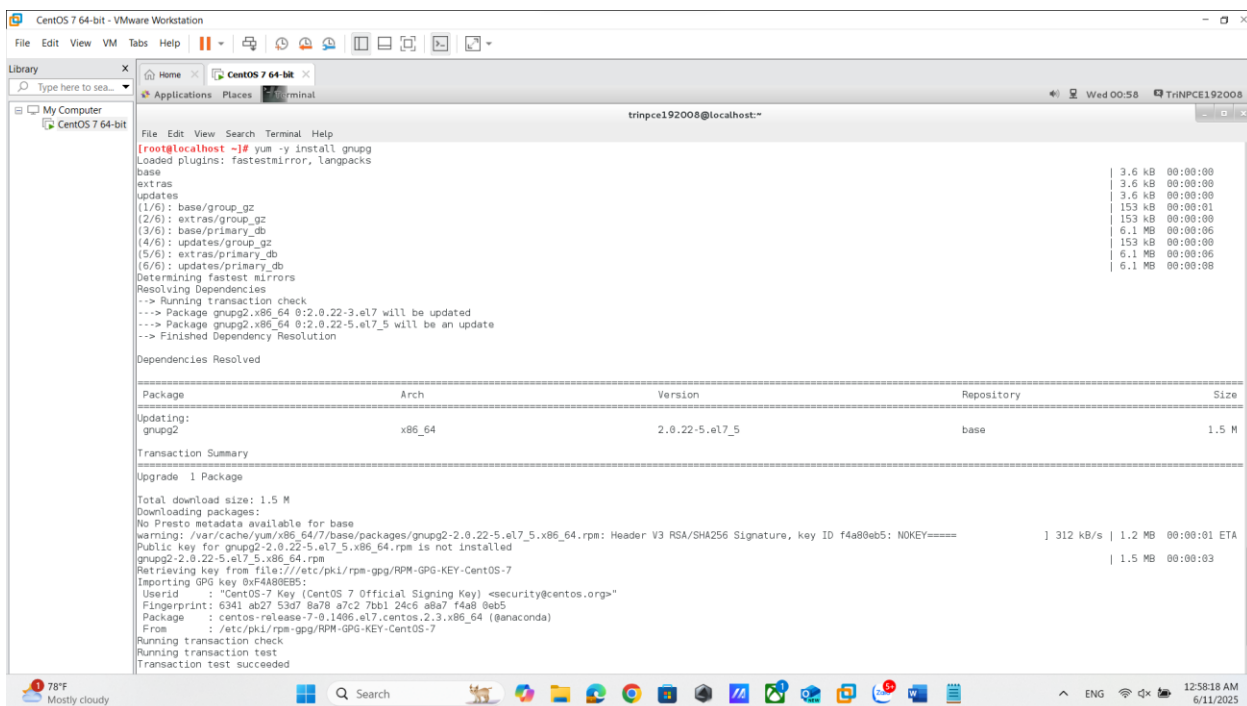
ảnh trên: load file sshkey.ppk lên putty



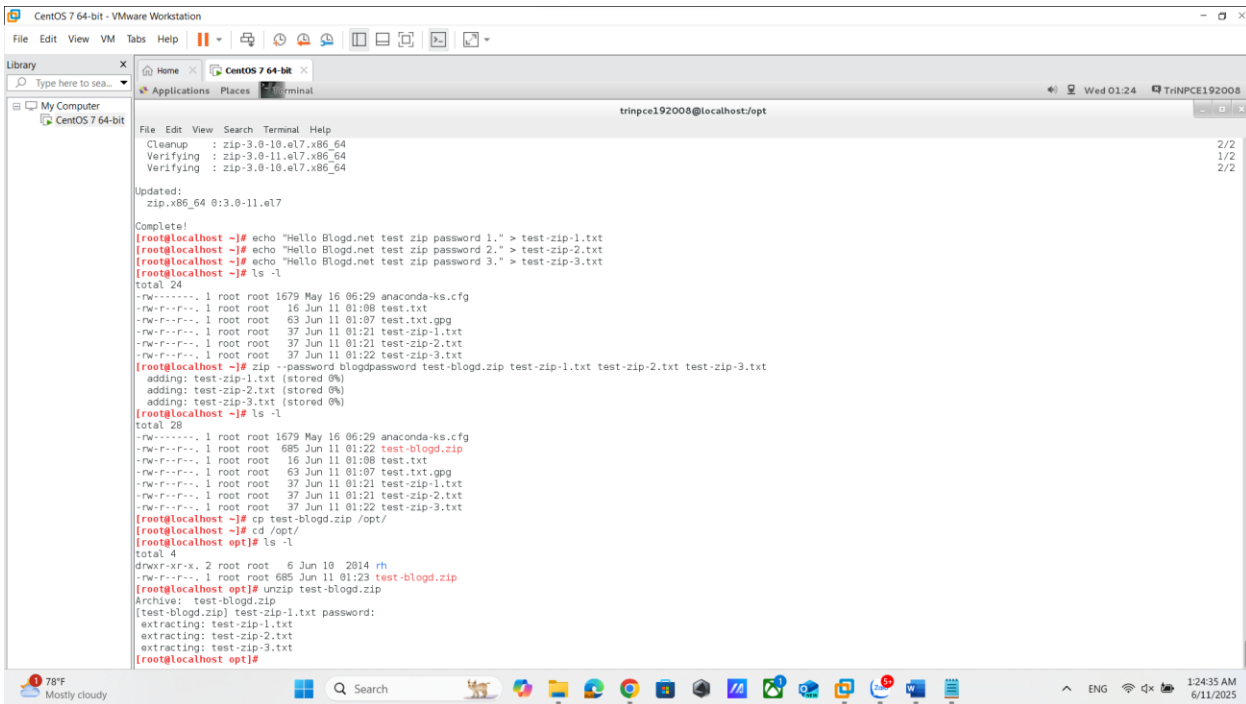
ảnh trên: kết nối ssh U1 thành công bằng file key

ENCRYPT:

1. GnuPG



ảnh trên: cài GnuPG



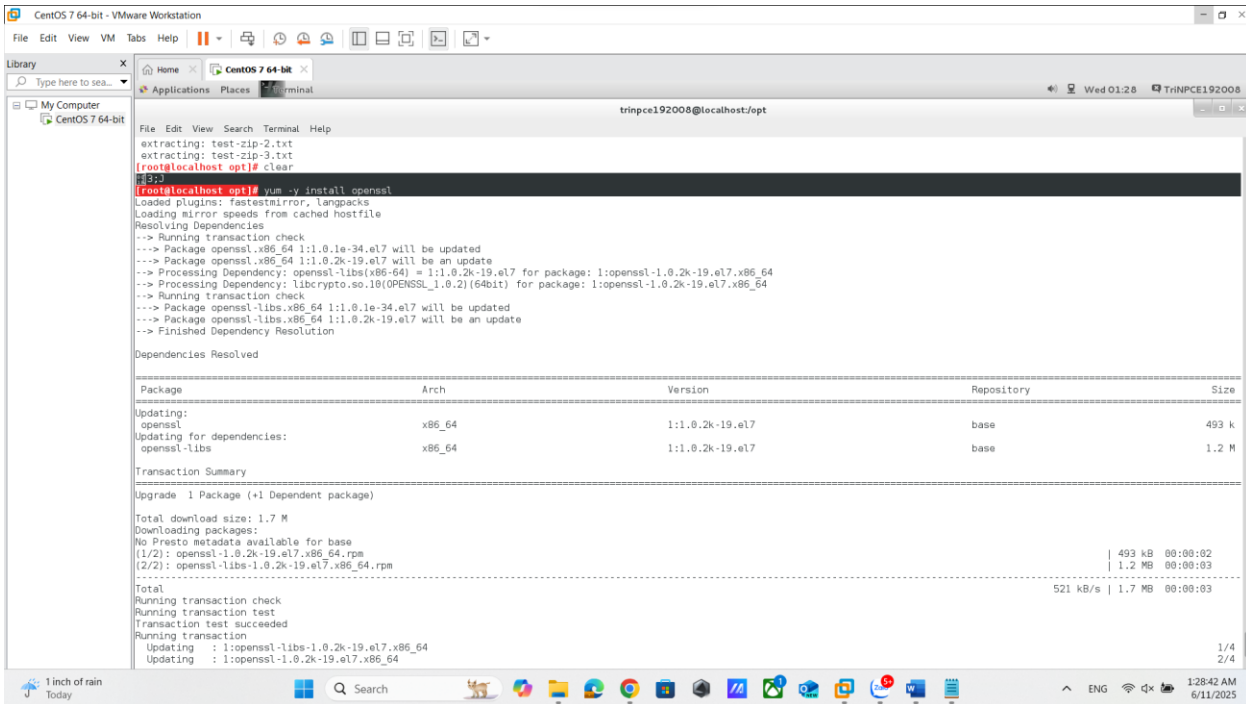
```
File Edit View Search Terminal Help
Cleanup : zip-3.0-10.el7.x86_64 2/2
Verifying : zip-3.0-11.el7.x86_64 1/2
Verifying : zip-3.0-10.el7.x86_64 2/2

Updated:
zip.x86_64 0:3.0-11.el7

Complete!
[root@localhost ~]# echo "Hello Blogd.net test zip password 1." > test-zip-1.txt
[root@localhost ~]# echo "Hello Blogd.net test zip password 2." > test-zip-2.txt
[root@localhost ~]# echo "Hello Blogd.net test zip password 3." > test-zip-3.txt
[root@localhost ~]# ls -l
total 24
-rw-r--r-- 1 root root 1679 May 16 06:29 anaconda-ks.cfg
-rw-r--r-- 1 root root 16 Jun 11 01:08 test.txt
-rw-r--r-- 1 root root 63 Jun 11 01:07 test.txt.gpg
-rw-r--r-- 1 root root 37 Jun 11 01:21 test-zip-1.txt
-rw-r--r-- 1 root root 37 Jun 11 01:21 test-zip-2.txt
-rw-r--r-- 1 root root 37 Jun 11 01:22 test-zip-3.txt
[root@localhost ~]# zip -password blogdpassword test-blogd.zip test-zip-1.txt test-zip-2.txt test-zip-3.txt
adding: test-zip-1.txt (stored 0%)
adding: test-zip-2.txt (stored 0%)
adding: test-zip-3.txt (stored 0%)
[root@localhost ~]# ls -l
total 28
-rw-r--r-- 1 root root 1679 May 16 06:29 anaconda-ks.cfg
-rw-r--r-- 1 root root 685 Jun 11 01:22 test-blogd.zip
-rw-r--r-- 1 root root 16 Jun 11 01:08 test.txt
-rw-r--r-- 1 root root 63 Jun 11 01:07 test.txt.gpg
-rw-r--r-- 1 root root 37 Jun 11 01:21 test-zip-1.txt
-rw-r--r-- 1 root root 37 Jun 11 01:21 test-zip-2.txt
-rw-r--r-- 1 root root 37 Jun 11 01:22 test-zip-3.txt
[root@localhost ~]# cp test-blogd.zip /opt/
[root@localhost ~]# cd /opt/
[root@localhost opt]# ls -l
total 4
drwxr-xr-x 2 root root 6 Jun 10 2014 rh
-rw-r--r-- 1 root root 685 Jun 11 01:23 test-blogd.zip
[root@localhost opt]# unzip test-blogd.zip
Archive: test-blogd.zip
  test-blogd.zip test-zip-1.txt password:
  extracting: test-zip-1.txt
  extracting: test-zip-2.txt
  extracting: test-zip-3.txt
[root@localhost opt]#
```

ảnh trên: mã hoá và giải mã bằng Zip, chúng ta tạo 3 file test-zip-1.txt, test-zip-2.txt, test-zip-3.txt và nén chúng lại thành test-blogd.zip với pass là blogdpassword. Sau đó để giải mã chúng ta copy test-blogd.zip vào thư mục /opt rồi dùng lệnh unzip để giải mã, nhập mật khẩu và chúng ta sẽ giải nén được 3 file trong /opt

5. Mã hóa bằng openssl



```
File Edit View Search Terminal Help
[root@localhost ~]# clear
[root@localhost opt]# yum -y install openssl
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
Resolving Dependencies
--> Running transaction check
--> Package openssl.x86_64 1:1.0.1e-34.el7 will be updated
--> Package openssl.x86_64 1:1.0.2k-19.el7 will be an update
--> Processing Dependency: openssl-libs(x86-64) = 1:1.0.2k-19.el7 for package: 1:openssl-1.0.2k-19.el7.x86_64
--> Processing Dependency: libcrypto.so.1.0(OPENSLL_1.0.2) (64bit) for package: 1:openssl-1.0.2k-19.el7.x86_64
--> Running transaction check
--> Package openssl-libs.x86_64 1:1.0.1e-34.el7 will be updated
--> Package openssl-libs.x86_64 1:1.0.2k-19.el7 will be an update
--> Finished Dependency Resolution

Dependencies Resolved

Package Arch Version Repository Size
-----
Updating:
openssl x86_64 1:1.0.2k-19.el7 base 493 k
Updating for dependencies:
openssl-libs x86_64 1:1.0.2k-19.el7 base 1.2 M

Transaction Summary
Upgrade 1 Package (+1 Dependent package)
Total download size: 1.7 M
Downloading packages:
No Presto metadata available for base
(1/2): openssl-1.0.2k-19.el7.x86_64.rpm | 493 kB 00:00:02
(2/2): openssl-libs-1.0.2k-19.el7.x86_64.rpm | 1.2 MB 00:00:03
-----
Total 521 kB/s | 1.7 MB 00:00:03
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
  Updating : 1:openssl-libs-1.0.2k-19.el7.x86_64 1/4
  Updating : 1:openssl-1.0.2k-19.el7.x86_64 2/4
```

ảnh trên: cài OpenSSL

